

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. Examination (EC/ME/CL/CE/IT/EE)

December 2011

PY 101 Engineering Physics

Date: 15.12.2011, Thursday

Time: 01:30 p.m. to 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 Choose the correct answer for the following questions.

[07]

- (a) What is the relative error for $v = 6.50 \text{ m/s} \pm 0.13 \text{ m/s}$?
- (a) 2% (b) 1%
(c) 5% (d) 25%
- (b) If $A = 2.0 \pm 0.2 \text{ m}$ and $B = 0.10 \pm 0.01 \text{ s}$, what is A/B ?
- (a) $20 \text{ m/s} \pm 4 \text{ m/s}$ (b) $20 \text{ m/s} \pm 20 \text{ m/s}$
(c) $20 \text{ m/s} \pm 10 \text{ m/s}$ (d) $20 \text{ m/s} \pm 5 \text{ m/s}$
- (c) If the intensity of a source of sound is 10^{-6} W/m^2 , calculate the intensity level in dB.
- (a) 50dB (b) 60dB
(c) 80dB (d) 100dB
- (d) An ultrasonic wave is sent from a ship towards the bottom of the sea. It is found that the time interval between the sending and receiving of the wave is 1.6 s . What is the depth of the sea, if the velocity of sound in the seawater is 1400 m/s ?
- (a) 1120 m (b) 560 m
(c) 2240 m (d) 1750 m
- (e) The propagation of sound in medium depend upon _____.
- (a) Amplitude (b) Frequency
(c) Wavelength (d) Properties of the medium
- (f) Sound travels fastest through _____.
- (a) Iron (b) water
(c) vacuum (d) air
- (g) Which of the following laser emits light of wavelength in visible range?
- (a) Nd:YAG laser (b) CO_2 laser
(c) Ruby laser (d) Both (a) and (b)

Q - 2 (a) Consider the experiment to measure the velocity of ultrasound by ultrasonic interferometer. What are the possible sources of systematic error in this experiment? [3]

- (b) What is Weber- Fechner law? Derive the equation for sensitiveness of ear. [4]
- (c) What are ultrasonic waves? Explain with neat circuit diagram, the generation of ultrasonic waves using magnetostriction oscillator. What are the advantages and disadvantages of this method? [7]

OR

- Q - 2 (a)** In a projectile motion experiment suppose you have the following series of measurements of the distance (X) travelled by the projectile: 30.3 cm, 32.5 cm, 29.7 cm, 28.4 cm, 31.3 cm, 27.8 cm, 33.1 cm, 32.5 cm, 30.8 cm and 31.5 cm. Calculate the standard deviation and standard error in the measurement of length. [3]
- (b) Give Sabine's formula and explain the parameters involved. A classroom has dimensions $20 \times 15 \times 5 \text{ m}^3$ and the reverberation time inside the room is 3.5 sec. Calculate the total absorption of its surface and average absorption coefficient. [4]
- (c) If the reverberation time is lower than the critical value, how will it affect the acoustical quality of a building? Explain any five remedies to maintain reverberation time at optimum level. [7]

- Q - 3 (a)** Define: (i) Pumping, (ii) Population inversion and (iii) Metastable state. [3]
- (b) State any four differences between step index fiber and graded index fiber. [4]
- (c) Explain the construction and working of a four level molecular laser, with energy level diagram. [7]

OR

- Q - 3 (a)** Distinguish between spontaneous emission and stimulated emission. [3]
- (b) Discuss any four advantage of optical fiber communication system over the conventional coaxial communication system. [4]
- (c) How holography differ from ordinary photography? Explain the method of construction & reconstruction of Hologram. [7]

SECTION - II

Q - 4 Choose the correct answer for the following questions. [07]

- (a) What is the frequency of the electromagnetic waves radiated from a Josephson junction, if the voltage drop at the junction is $8.5 \mu V$? (Use $h = 6.62 \times 10^{-34} \text{ Js}$)
- (a) 4 GHz (b) 5 GHz
- (c) 6 GHz (d) 7 GHz
- (b) A superconductor is a perfect _____ material.
- (a) Diamagnetic (b) Dielectric
- (c) Insulator (d) none
- (c) The packing factor of BCC structure is _____.
- (a) 68 % (b) 52 %

- (c) 74 % (d) 46 %
- (d) Monochromatic X-rays of $\lambda = 1.5 \text{ \AA}$ are incident on a crystal face having an interplanar spacing of $1.6 \text{ \AA}$. What is the highest order for which Bragg's reflection can be seen?
- (a) 1 (b) 2
(c) 3 (d) 4
- (e) The crystal can serve as a _____ for X-ray diffraction.
- (a) 3-D grating (b) 1-D grating
(c) 2-D grating (d) none of the above
- (f) Cubic crystal system is represented by;
- (a) $a = b = c, \alpha = \beta = \gamma = 90^\circ$ (b) $a \neq b \neq c, \alpha = \beta = \gamma = 90^\circ$
(c) $a = b \neq c, \alpha = \beta = \gamma = 90^\circ$ (d) $a = b = c, \alpha = \beta = \gamma \neq 90^\circ$
- (g) The surface area to volume ratio is _____ for nanomaterials.
- (a) very large (b) very less
(c) moderate (d) none of the above

- Q - 5 (a) List out the differences between a.c. Josephson effect and d.c. Josephson effect. [3]
(b) The Bragg's angle for the 1st order reflection from the (111) plane in a FCC crystal for an X-ray wavelength of $1.54 \text{ \AA}$ is 19.2° . Compute the cube edge of the unit cell. [4]
(c) What is Hall effect? Derive an expression for Hall voltage, Hall coefficient, and Hall mobility. [7]

OR

- Q - 5 (a) Based on the energy band model classify the solids. [3]
(b) The cut off wavelength from an X-ray tube operating at 50 kV is diffracted from NaCl crystal. The first order maximum is observed at a Bragg angle of 26° . Calculate the interplanar spacing and the angle through which the receiver of the diffractometer must be rotated to observe for the 2nd order maxima. [4]
(c) What are the characteristics of X-rays? How are X-rays generated? [7]
- Q - 6 (a) Sketch the following Miller indices planes of a cubic cell: (001), (010) and (222) [3]
(b) Explain the unit cell and lattice parameters. What are a primitive cell and a non-primitive cell? [4]
(c) What are nanomaterials? Write any five important applications of nanomaterials. [7]

OR

- Q - 6 (a) Distinguish between crystalline & non-crystalline solids. [3]
(b) Calculate the interplanar spacing between (111) and (321) planes in FCC crystal. Atomic radius is given as $1.426 \text{ \AA}$. [4]
(c) Why nano material exhibit different properties? Explain. [7]
